NMC 2014 O-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn. 1. Simplify the following expression $\frac{(-a)^3-(\frac{1}{a})^{-2}}{-a^2+(-\frac{1}{a})^{-3}}$.

Qn. 2. Determine the number of pairs (x, y) of integer solutions for $2^{2x} - 3^{2y} = 55$.

Qn. 3. A number system based on 26 uses the letters of the alphabet as its digits, with $A = 0, B = 1, C = 2, ..., Y = 24$, and $Z = 25$. Express $ONE + ONE$ in this system.

Qn. 4. The harmonic mean of two numbers is defined to be the reciprocal of the average of the reciprocals of the numbers. Find x such that the harmonic mean of 4 and x is 6.

Qn. 5. The ratio of children to adults at a party is 2 : 3. A busload of 30 more children arrives at the party, and now the ratio of children to adults is 3 : 2. How many people were at the party before the bus arrived?

Qn. 6. It takes 852 digits to number the pages of a book consecutively. How many pages are there in the book?

Qn. 7. Two athletes **A** and **B** exercise along the same route. **B** jogs the first half of the route at $6km/hr$ and runs the remainder of the route at $12km/hr$ taking a total time of x hours. **A** walks the first third of the route at $5km/hr$ and runs the remainder of the route at $15km/hr$ taking a total time of y hours. Determine the value of $\frac{x}{y}$.

Qn. 8. If $TEEN$ is a four-digit number such that $TEEN_5 + TEEN_7 = TEEN_8$, what is the value of the digit N?

Qn. 9. Toshiba laptops are packed either in boxes of 5 or boxes of 12. What is the smallest number of full boxes required to pack exactly 2005 Toshiba laptops?

Qn.10. PQR is a triangle such that $PQ = 19cm$, $QR = 18cm$ and $PR = 17cm$. Let S be a point on PQ, T be a point on PR and U be a point on ST such that $QS = SU$ and $UT = TR$. Find the perimeter of triangle PST.

SECTION B

Qn.11. If m and n are non-negative real numbers such that $m - 2n > 0$ and

$$2\log(m - 2n) = \log m + \log n.$$

Find the ratio of m to n.

Qn.12. Alex and John together can do job in 2 days. John and Jimmy can the same job in four days while Alex and Jimmy can do the job in $2\frac{2}{5}$ days. Find the number of days required for Alex to do the job alone.

Qn.13. A fruit salad consists of blueberries, raspberries, grapes, and cherries. The fruit salad has a total of 280 pieces of fruit. There are twice as many raspberries as blueberries, three times as many grapes as cherries, and four times as many cherries as raspberries. How many cherries are there in the fruit salad?

Qn.14. In rectangle $ABCD$, $AB = 5cm$ and $BC = 3cm$. Points F and G are on CD with $DF = 1cm$ and $CG = 2cm$, and lines AF and BG intersect at E. What is the area of triangle AEB?

Qn.15. The odd numbers from 1 to 17 can be used to build a 3×3 magic square (the rows and columns have the same sum). If the $1, 5$ and 13 are as shown, what is x?

	1	
5		13
x		

NMC 2014 O-LEVEL PAPER 1 SOLUTIONS

SECTION A

1.

$$\frac{(-a)^3-(\frac{1}{a})^{-2}}{-a^2+(-\frac{1}{a})^{-3}}=\frac{(-a)^3-\frac{1}{(\frac{1}{a})^2}}{-a^2+\frac{1}{(-\frac{1}{a})^3}}=\frac{-a^3-a^2}{-a^2-a^3}=\frac{-a^3-a^2}{-a^3-a^2}=1.$$

2. The left hand side can be factorized to give $(2^x-3^y)(2^x+3^y)=55$ so either $2^x-3^y=1$ and $2^x+3^y=55$ or $2^x-3^y=5$ and $2^x+3^y=11$. Solving simultaneously the first two equations gives $2^x=28$ and $3^y=27$, which does not have an integer solution for x. The second possibility gives $2^x=8$ and $3^y=3$, which has one integer solution $(x,y)=(3,1)$.

3. $BDAI$. The rightmost position is $4+4=8=I$. The next position is $13+13$, for which we enter $0=A$ and carry 1. The next position is $1+14+14$, for which we enter $3=D$ and carry $1=B$ to the leftmost position.

4. Let the numbers be x and y.

$$\begin{aligned}\text{Harmonic mean of } x \text{ and } y &= \frac{1}{\frac{1}{2}(\frac{1}{x}+\frac{1}{y})}=\frac{4xy}{2x+2y}=\frac{2xy}{x+y}\\ 6 &= \frac{2\times x\times 4}{x+4}\\ 6x+24 &= 8x\\ x &= 12.\end{aligned}$$

5. Let A represent the number of adults while C represent the number of Children. Then
   Initially, $\frac{C}{A}=\frac{2}{3}\Longrightarrow C=\frac{2}{3}A$.
   After the bus load, we have $\frac{C+30}{A}=\frac{3}{2}\Longrightarrow C+30=\frac{3}{2}A$.
   Substituting for C in the equation gives

$$\frac{2}{3}A+30=\frac{3}{2}A$$

$$\frac{5A}{6}=30\Longrightarrow A=36.$$

Therefore $C = \frac{2}{3} \times 36 = 2 \times 12 = 24$. Thus the number of people at the party before the bus arrived are $24 + 36 = 60$.

6. From pages 1 through 9, we use 9 digits. From page 10 through 99, we use $((99-10)+1) \times 2 = 90 \times 2 = 180$ digits which makes a total of $9 + 180 = 189$ digits for pages 1 through 99. That leaves $852 - 189 = 663$ digits remaining to be used. These are obtained by going $\frac{663}{3} = 221$ pages beyond page 99 which makes up to page $99 + 221 = 320$. Therefore there are 320 pages of the book.

7. Let the length of the route be d km. Then B jogs $\frac{d}{2}$ km at 6 km/h and runs $\frac{d}{2}$ km at 12 km/h. Since the total time taken was x hours, then

$$x = \frac{\text{Distance}}{\text{Speed}} = \frac{\frac{d}{2}}{6} + \frac{\frac{d}{2}}{12} = \frac{d}{12} + \frac{d}{24} = \frac{2d+d}{24} = \frac{3d}{24} = \frac{d}{8}.$$

A walks $\frac{d}{3}$ km at 5 km/h and runs for $\frac{2}{3}$ km at 15 km/h.

$$y = \frac{\text{Distance}}{\text{Speed}} = \frac{\frac{d}{3}}{5} + \frac{\frac{2d}{3}}{15} = \frac{d}{15} + \frac{2d}{45} = \frac{d}{9}.$$

Thus

$$\frac{x}{y} = \frac{d}{8} \div \frac{d}{9} = \frac{d}{8} \times \frac{9}{d} = \frac{9}{8}.$$

8. Note that all three digits must be either $0, 1, 2, 3$ or 4. Further more,
$TEEN_5 = 125T + 25E + 5E + N = 125T + 30E + N$
$TEEN_7 = 343T + 49E + 7E + N = 343T + 56E + N$
$TEEN_8 = 512T + 64E + 8E + N = 512T + 72E + N$
Therefore, $125T + 30E + N + 343T + 56E + N = 512T + 72E + N$, from which $44T - 14E = N$. Suppose $T = 1$, then $E = 3$ and $N = 2$. If $T = 2, 3$ or 4, then N must be greater than 4 for E to be less than 5 which would be a contradiction. Hence $N = 2$.

9. In order to have the smallest number of full boxes, we must have as many boxes of 12 laptops as possible. Suppose the laptops are packed in the boxes of 12, then 167 boxes can be used and 1 laptop remains. This is not the right choice as 1 laptop cannot fill any box. If 166 of 12 are used, then $1 + 12 = 13$ laptops will remain and these cannot fill up the entire box. If 165 of 12 are used, then $1 + 12 + 12 = 25$ laptops will remain and these can fill up to 5 boxes that contain 5 laptops each. Thus, the minimum number of full boxes are 165 big boxes + 5 small boxes = 170 full boxes.

10. Perimeter of $\triangle PST = PS + ST + PT = PS + (SU + UT) + PT = PS + SQ + TR + PT = (PS + SQ) + (PT + TR) = PQ + PR = 19cm + 17cm = 36cm$.

SECTION B

11.

$$\begin{aligned} 2\log(m-2n) &= \log m + \log n \\ \log(m-2n)^2 &= \log mn \Longrightarrow m^2 - 4mn + 4n^2 = mn \end{aligned}$$

$$\frac{m^2}{n^2} - 5\frac{m}{n} + 4 = 0 \Longrightarrow \left(\frac{m}{n}\right)^2 - 5\left(\frac{m}{n}\right) + 4 = 0.$$

$$y^2 - 5y + 4 = 0 \text{ where } \frac{m}{n} = y.$$

$$y = \frac{+5 \pm \sqrt{25-16}}{2} = \frac{5 \pm 3}{2} = 4 \text{ or } 1.$$

Suppose $y = 1$, then $\frac{m}{n} = 1 \Longrightarrow m = n$. This means that $m - 2n = n - 2n = -n$ which contradicts the fact that m and n are non-negative real numbers. Therefore y cannot be equal to 1. Suppose $y = 4$, then $\frac{m}{n} = 4 \Longrightarrow m = 4n$. Hence $m - 2n = 4n - 2n = 2n > 0$. Hence the ratio $m : n = 4 : 1$.

12. Let a, b, c denote the respective number of days it would take Alex, John and Jimmy to complete the job if he was working alone. Then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ represents the fractions of the job done by each in one day. Since it takes Alex and John together 2 days to do the whole job, it implies they both do half the job in one day. We thus have the following equations.

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{2} \tag{1}$$

$$\frac{1}{b} + \frac{1}{c} = \frac{1}{4} \tag{2}$$

$$\frac{1}{a} + \frac{1}{c} = \frac{5}{12} \tag{3}$$

Subtracting Equation (2) from Equation (1) gives $\frac{1}{a} - \frac{1}{c} = \frac{1}{4}$. This added to Equation (3) gives $\frac{2}{a} = \frac{8}{12} \Longrightarrow 8a = 24$ so that $a = 3$. Therefore it would take Alex 3 days to do the job alone.

13. For each blueberry in the fruit salad there are 2 raspberries, 8 cherries and 24 grapes. Thus there are $1 + 2 + 8 + 24 = 35$ pieces of fruit for each blueberry. Because $280 = 35 \times 8$, it follows that there are a total of 8 blueberries, $8 \times 2 = 16$ raspberries, $8 \times 8 = 64$ cherries, and $8 \times 24 = 192$ grapes in the fruit salad. Thus there are 64 cherries.

14. Let H be on the line AB such that EH is perpendicular to AB and I be the intersection of EH and DC. Then $FG = 2cm$. Since $\triangle ABE$ is similar to $\triangle FGE$, we have that $\frac{FG}{AB} = \frac{EI}{EH} = \frac{EI}{EI+IH}$. Thus $\frac{2}{5} = \frac{EI}{EI+3}$ so that $EI = 2$. Hence $EH = EI + IH = 2 + 3 = 5$. The area of

    $\triangle ABE = \frac{1}{2}AB \times EH = \frac{1}{2} \times 5 \times 5 = \frac{25}{2} = 12.5cm^2$.

15. The sum of odd numbers from 1 to 17 is $1+3+5+7+9+11+13+15+17 = 81$. Since the rows and columns have the same sum, the sum in any row or column is $\frac{81}{3} = 27$.

	1	
5	p	13
x	q	r

    From the magic square, we have

$$\begin{aligned} 5+p+13 &= 27 \Longrightarrow p = 27-18 = 9. \\ 1+p+q &= 27 \Longrightarrow q = 27-1-9 = 17. \\ x+q+r &= 27 \Longrightarrow x+r = 27-17 = 10. \end{aligned}$$

    Since $x + r = 10$, then either $x = 3$ or $r = 7$ or $x = 7$ and $r = 3$. Suppose $x = 3$, then we would have to use 7 in the bottom right square which would require another 7 in the top right square. This eliminates $x = 3$ since it would require that we use 7 twice in the magic square. Therefore the value of x is 7.